EMAIL SPAM DETECTION

DATABASE MANAGEMENT SYSTEM

Review 3 – Methodology Development

Group No.: 13

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PROBLEM STATEMENT & OBJECTIVES

THE CHALLENGE

Spam emails drive phishing, fraud, and unwanted content

High email volumes demand organized storage and fast retrieval

A structured database is essential to manage users, emails, and spam information efficiently

OUR OBJECTIVES

Design a normalized relational database

Store and manage user and email data

Maintain spam-detection records

Enable efficient search and retrieval

Apply DBMS features: keys, joins, indexing, triggers, stored procedures



PROPOSED METHODOLOGY & WORKFLOW

USER MANAGEMENT

↓

EMAIL MANAGEMENT

↓

SPAM DETECTION

↓

SEARCH & CLASSIFICATION

↓

DATABASE TESTING & EVALUATION

METHODOLOGY

Store users in relational database

Store emails with sender/receiver relationships

Link emails with spam-detection info

Use SQL queries and JOIN operations for retrieval

Assign spam scores and classifications

Test system with sample emails

Extend system with advanced DBMS features



DATABASE DESIGN & TECHNOLOGY

USERS

user_id (PK)
name
email
password
created_at

→

EMAILS

email_id (PK)
sender_id (FK)
receiver_id (FK)
subject
body
is_read
created_at

→

SPAM DETECTION

detection_id (PK)
email_id (FK)
spam_score
classification
detected_at

RELATIONSHIPS

Users → Emails → Spam Detection

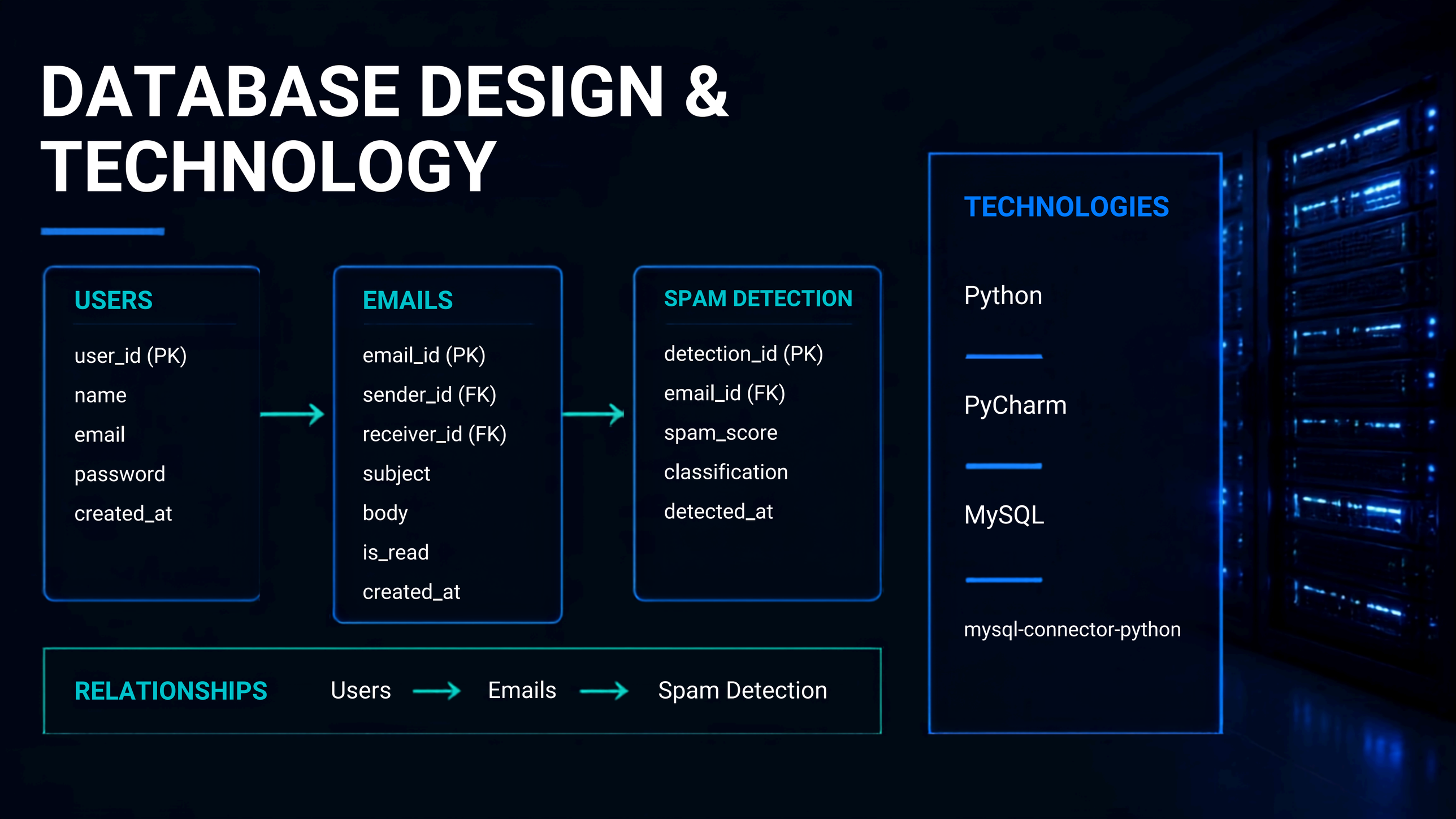
TECHNOLOGIES

Python

PyCharm

MySQL

mysql-connector-python



IMPLEMENTATION PROGRESS

COMPLETED

MySQL server setup

email_spam_detection database

Core relational tables

Primary and foreign keys

User and email data management

Sender-receiver relationships

Initial spam-detection records

SQL SELECT, UPDATE, and JOIN testing

IN PROGRESS

Python-MySQL integration

Email search and classification

interface

Indexing for performance

Triggers and automation

Stored procedures

Larger dataset testing



INTERMEDIATE RESULTS

WHAT WE STORED AND FOUND

Users and emails stored successfully

Sender-receiver relationships maintained

Spam-detection linked to email records

2

USERS

Alice and Danny

2

EMAILS

Test messages

2

CLASSIFICATIONS

Spam and not spam

MYSQL RESULT SNAPSHOTS

Users Table

```
mysql> DESCRIBE users;
+------------+--------------+------+-----+-------------------+----------------+
| Field      | Type         | Null | Key | Default           | Extra          |
+------------+--------------+------+-----+-------------------+----------------+
| user_id    | int(11)      | NO   | PRI | NULL              | auto_increment |
| name       | varchar(100) | NO   |     | NULL              |                |
| email      | varchar(150) | NO   | UNI | NULL              |                |
| password   | varchar(255) | NO   |     | NULL              |                |
| created_at | timestamp    | NO   |     | CURRENT_TIMESTAMP |                |
+------------+--------------+------+-----+-------------------+----------------+
5 rows in set (0.01 sec)
```

Emails Table

```
mysql> DESCRIBE emails;
+-------------+--------------+------+-----+-------------------+----------------+
| Field       | Type         | Null | Key | Default           | Extra          |
+-------------+--------------+------+-----+-------------------+----------------+
| email_id    | int(11)      | NO   | PRI | NULL              | auto_increment |
| sender_id   | int(11)      | NO   | MUL | NULL              |                |
| receiver_id | int(11)      | NO   | MUL | NULL              |                |
| subject     | varchar(255) | NO   |     | NULL              |                |
| body        | text         | NO   |     | NULL              |                |
| is_read     | tinyint(1)   | YES  |     | 0                 |                |
| created_at  | timestamp    | NO   |     | CURRENT_TIMESTAMP |                |
+-------------+--------------+------+-----+-------------------+----------------+
7 rows in set (0.00 sec)
```

Spam_Records Table

```
mysql> SELECT * FROM spam_detection;
+--------------+----------+------------+----------------+---------------------+
| detection_id | email_id | spam_score | classification | detected_at         |
+--------------+----------+------------+----------------+---------------------+
|            1 |        1 |         95 | SPAM           | 2026-09-15 17:09:24 |
|            2 |        2 |          5 | NOT SPAM       | 2026-09-15 17:09:24 |
+--------------+----------+------------+----------------+---------------------+
2 rows in set (0.00 sec)
```

SQL JOIN & SPAM DETECTION DEMONSTRATION

WHAT THIS SHOWS

- Demonstrates foreign-key relationships
- Shows multi-table retrieval
- Practical SQL JOIN operations
- Connects email records with spam-detection results

SQL JOIN QUERY

```
mysql> SELECT
    ->     u1.name AS sender,
    ->     u2.name AS receiver,
    ->     e.subject,
    ->     e.body,
    ->     s.spam_score,
    ->     s.classification
    -> FROM emails e
    -> JOIN users u1
    ->     ON e.sender_id = u1.user_id
    -> JOIN users u2
    ->     ON e.receiver_id = u2.user_id
    -> JOIN spam_detection s
    ->     ON e.email_id = s.email_id
    -> ORDER BY s.spam_score DESC;
```

EXAMPLE OUTPUT

```
+----------+--------+----------+---------------------------------------+------------+----------------+
| email_id | sender | receiver | subject                               | spam_score | classification |
+----------+--------+----------+---------------------------------------+------------+----------------+
|        1 | Danny  | Alice    | Congratulations! You won a FREE prize |         95 | SPAM           |
|        2 | Alice  | Danny    | DBMS Project Meeting                  |          5 | NOT SPAM       |
+----------+--------+----------+---------------------------------------+------------+----------------+
2 rows in set (0.01 sec)
```

NOVELTY, ADAPTABILITY & EVALUATION

NOVELTY

Brings email storage, management, retrieval, and spam detection together in one relational DBMS

Combines user, email, and spam detection

Maintains links between emails and detection results

ADAPTABILITY

Easily add new users and emails

Extend spam-detection attributes

Update detection rules/models as patterns change

Handles larger datasets with scalability

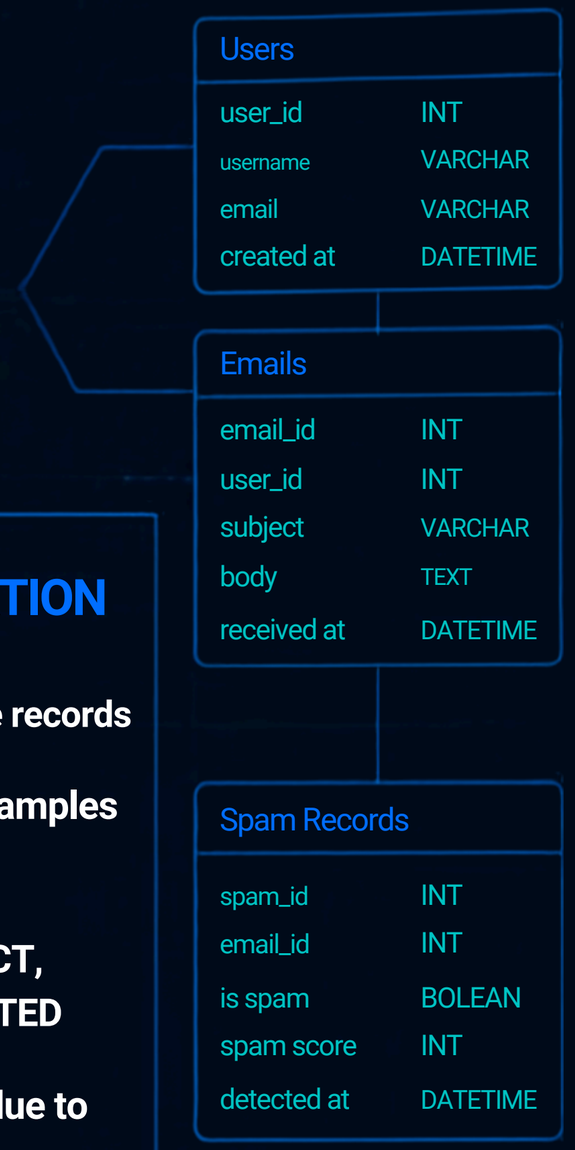
CURRENT EVALUATION

Initial testing with sample records

Spam and non-spam examples are represented

INSERT, UPDATE, SELECT, JOIN OPERATIONS TESTED

Preliminary evaluation due to small dataset



CONCLUSION & NEXT STEPS

WHERE WE STAND

Database configured successfully

Core relational tables implemented

Users and emails stored

Primary and foreign keys implemented

Initial span-detection prototype created

SQL JOIN-based retrieval demonstrated

WHAT'S NEXT

Complete Python-MySQL integration

Implement email search and classification

Add indexes, triggers, stored procedures

Improve spam classification

Expand dataset

Perform systematic testing and evaluation

Integrate all modules into final system

EXPECTED OUTCOME

A normalized, efficient, reliable DBMS for managing email and spam-related information.

